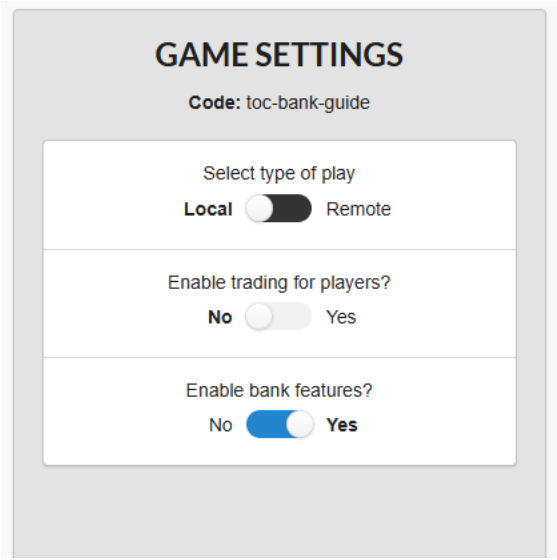
# Tragedy of the Commons

Facilitation Materials – Central Bank Guide

The central bank is an optional game mechanic in ToC that adds new functionality and can allow facilitators greater control over the flow of currency (i.e. fish) in the game. Since it is optional, facilitators have the freedom to decide how they want to use the bank, if at all. For facilitators who are interested in utilizing the feature, this **Central Bank Guide** will give a detailed explanation on how to set up the central bank and incorporate it into a game.

This guide is intended for teachers who have already read the Introductory Guide and Step-by-Step Guide.




## Setup

After creating the game, the facilitator will see the Game Settings panel on the setup dashboard.

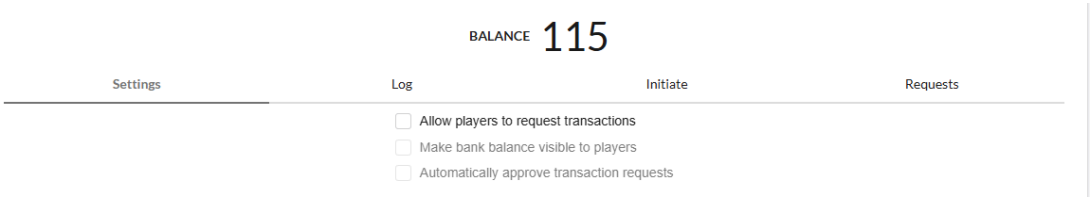
- The bank features can be enabled by setting the **Enable bank features** toggle to Yes.
- This toggle can be changed at any time by returning to the setup dashboard, so the facilitator can enable or disable it while the game is ongoing if desired.

While the trading mechanic is similar and/or related to the bank, it does not need to be enabled for the bank to function, and vice versa.

## During the Game

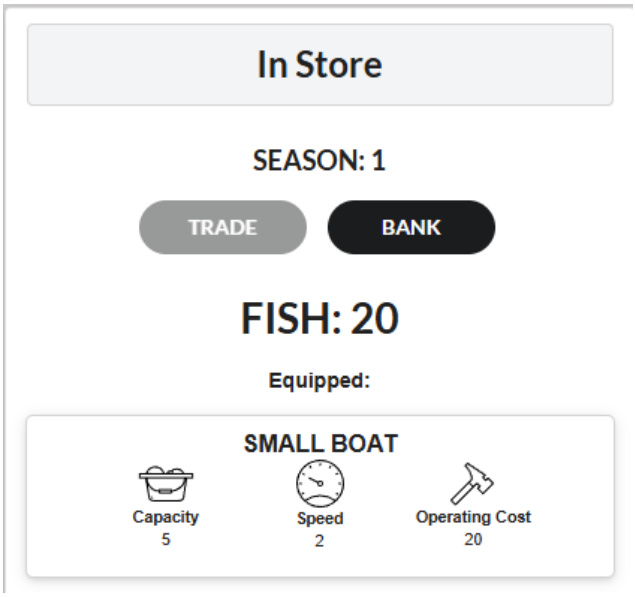
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| **Bank** |  In the game dashboard, facilitators will see the four tabs pictured above on the game dashboard. The bank features can be accessed under the Bank tab. By default, the bank starts with a balance of zero. If a nonzero initial balance is desired, this must be manually set up by the facilitator using the **Initiate** feature, as explained below.<br><br>Beneath the Bank tab are four more sub-tabs, as pictured and explained below. |

## Settings

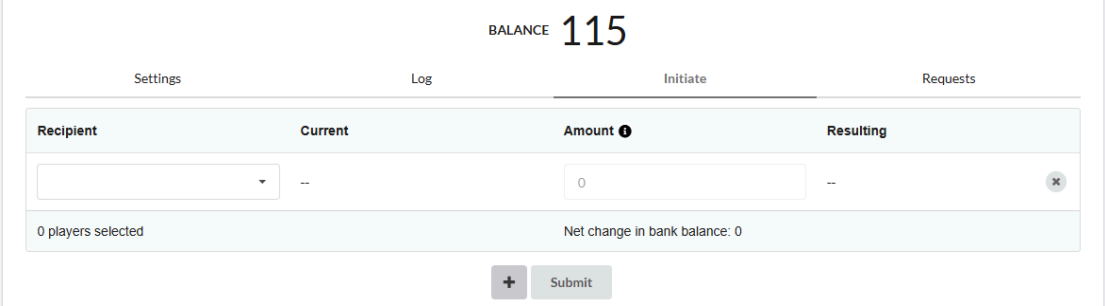



Settings has three options that can be toggled:

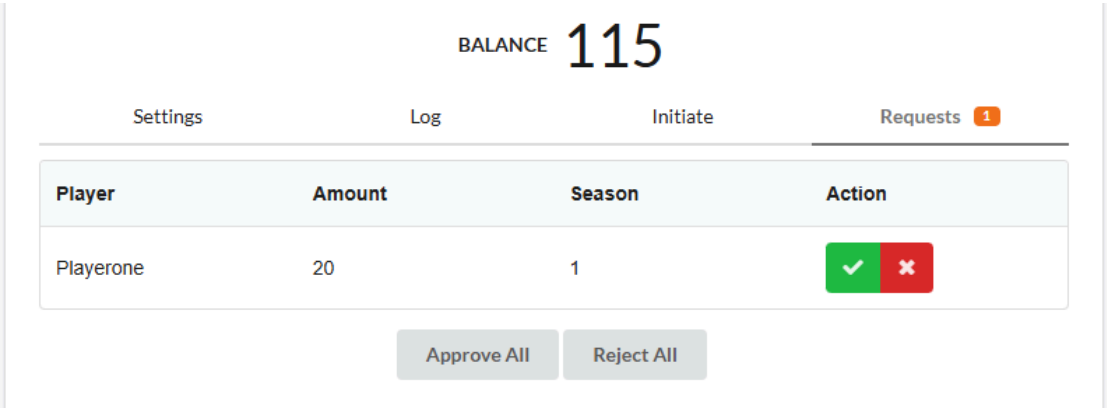
- Allow players to request transactions:
  - When enabled, players will be able to request to give fish to or receive fish from the bank (i.e., make a transaction) on their screens, but only when they are in the store. These requests can be viewed and handled in the **Requests** tab.
  - If enabled, the player screen will look like this, where they can press the **Bank** button to request a transaction:



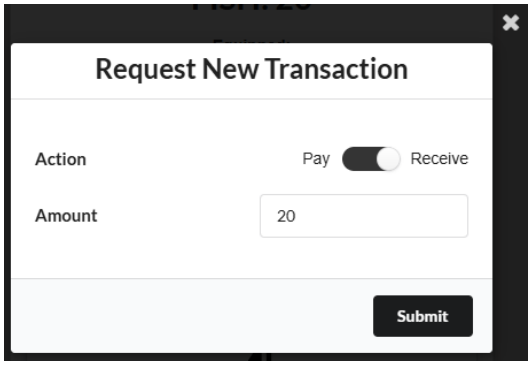

- Make bank balance visible to players:
  - When enabled, the bank's balance is visible to players when they make a transaction request.
  - This can only be enabled if "Allow players to request transactions" is enabled.
- Automatically approve transaction requests:
  - When enabled, any requests players make or have made are automatically approved.
  - This can only be enabled if "Allow players to request transactions" is enabled.

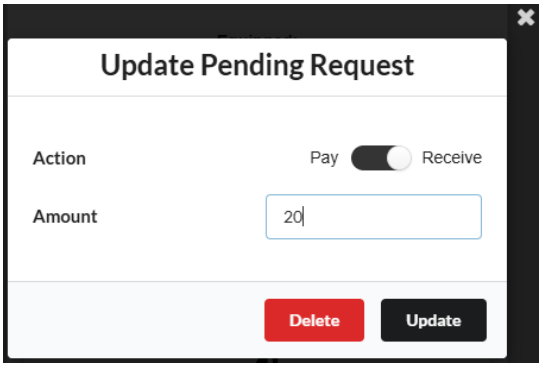
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| **Log** | The log is a record of all transactions to/from the bank. It displays the:<br><br>● Recipient type<br>  ○ This can either be the player or bank.<br>● Name of the recipient<br>  ○ This can either The Bank, All Players, or the username of an individual player.<br>● Amount of fish received by that recipient<br>  ○ A positive amount means the bank paid that amount to the recipient.<br>  ○ A negative amount means the bank received that amount from the recipient.<br>● Season in which the transaction occurred |
| **Initiate** | In this sub-tab, the facilitator can make transactions with individual players, all players, or the bank itself (which directly adjusts the bank balance).<br>  <br>● Initiate a transaction by pressing the plus button at the bottom. This brings up a new row in the table above. After selecting the recipient and amount, press **Submit** to apply these changes.<br>  ○ If a transaction should not be completed, pressing the X on the right side of a row will cancel that transaction.<br>● When the recipient is a player (or All Players), a positive amount moves fish from the bank to the player, increasing the player's fish balance by the specified amount and decreasing the bank's balance by the same amount. A negative amount does the opposite, taking fish from the player.<br>  ○ If the recipient is an individual player, the Current and Resulting columns show the current and resulting amounts of fish for the recipient after the transaction.<br>  ○ Selecting All Players as the recipient transfers the specified amount of fish to/from each player. For example, for a transaction with 5 as the amount and All Players as the recipient, the bank gives 5 fish to every player.<br>  ■ If there is an unsubmitted transaction with a player, it is impossible to initiate a transaction with All Players, as All Players will not show up in the list of recipients. (Likewise, it is impossible to initiate two transactions at the same time with any individual player).<br>● Selecting The Bank as the recipient adjusts the bank's balance directly.<br>  ○ A positive amount increases the bank's balance by that much, and a negative amount decreases the bank's balance. |

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| **Requests** | • This sub-tab lists all the incoming requests that players have made and allows the facilitator to manage them. Note that this list will be empty if players are not allowed to make requests or their requests are auto-approved. |




- Each request shows which player is making it, what amount they've requested, which season they made the request in, and an **Action** option.
  - The **Action** is where the facilitator can approve (green checkmark) or reject a request (red X).
  - If the amount is negative, that means the player is requesting to pay the bank that amount, while if the amount is positive, the player is requesting to receive that amount from the bank.
- Use the **Approve All** or **Reject All** buttons at the bottom of the request list to quickly manage many requests.
- Students can only make one request at a time, and they can update their pending requests to change the amount.
  - Player screens while creating and updating requests will look similar to these:



# After the Game

As explained in the Discussion Guide, there should be a student-driven class discussion after the first playthrough of the game, in which students should share their thoughts and experiences and come up with new experiments and strategies for how to preserve the fish populations in their next playthrough of the game.

- The bank can and should be encouraged to be part of these experimental strategies. Furthermore, the bank can be used to introduce more economic concepts (such as taxation, insurance, loans, etc.) to the players and put more of a social science focus on the game.
- For example:
    - Players could take loans from the bank to buy better equipment
    - Players could pay seasonal taxes to the bank in exchange for insurance if they are in debt or close to game over

There are many ways to incorporate it into your class's playthroughs of the game in order to best get across the concepts you want to teach, and facilitators can take a lot of creative liberties with how they incorporate the bank.

If you would like to share how you used this feature in your classroom play, please send us a note to psims@mit.edu!